

Asymptotic statistics on stratified spaces

Lecture 2: CLTs on metric spaces and manifolds

Slides available at github.com/statdivlab/asymptotics-stratified-spaces

Amy Willis

September 2026

Recap

Last lecture:

- ▶ CLT for averages in $\mathbb{R}^d$:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{\mathcal{D}} N_d(0, \Sigma),$$

- ▶ A CLT for minimizers $\theta_0 \in \Theta \subseteq \mathbb{R}^s$ open of a function m

$$\theta_0 = \arg \min_{\theta \in \Theta} \int m(\theta, X) dP = \arg \min_{\theta \in \Theta} P m_\theta$$

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n m(\theta, X_i) = \arg \min_{\theta \in \Theta} P_n m_\theta$$

Theorem: a CLT for M-estimators

Theorem: Assume (M1)–(M6). Then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{\mathcal{D}} N_s(0, \Lambda^{-1} \Sigma \Lambda^{-1}),$$

where

$$\Sigma = P \dot{m}_{\theta_0} \dot{m}_{\theta_0}^T = \text{Var}(\dot{m}_{\theta_0}(X_1)).$$

and

$$\Lambda = P \ddot{m}_{\theta_0}.$$

Theorem: a CLT for M-estimators

WTS: $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{\mathcal{D}} N_s(0, \Lambda^{-1}\Sigma\Lambda^{-1})$.

Proof overview:

1. $P_n \dot{m}_{\hat{\theta}_n} = 0$ on a set with probability $\rightarrow 1$
2. $P_n \dot{m}_{\hat{\theta}_n} = P_n \dot{m}_{\theta_0} + \Lambda_n h_n$ for $h_n = \hat{\theta}_n - \theta_0$ and some Λ_n
3. $\Lambda_n \xrightarrow{P} \Lambda$
4. $h_n = -\Lambda_n^{-1} P_n \dot{m}_{\theta_0} = -(\Lambda^{-1} + o_p(1)) P_n \dot{m}_{\theta_0}$
5. $\sqrt{n}(\hat{\theta}_n - \theta_0) = -\sqrt{n}\Lambda^{-1} P_n \dot{m}_{\theta_0} + o_p(1)$
6. Apply CLT: $\sqrt{n} P_n \dot{m}_{\theta_0} \xrightarrow{\mathcal{D}} N(0, \Sigma)$

More details are given in the Lecture 1 slides.

Generalizing averages to $\mathcal{X}, \Theta \neq \mathbb{R}^d$

Averages and minimizers in $\mathbb{R}^d$ have normal limiting laws - great!

Let's turn to non- $\mathbb{R}^d$ settings...

Generalizing averages to $\mathcal{X}, \Theta \neq \mathbb{R}^d$

Averaging a set of observations $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ relies on two essential vector space operations: addition and scalar multiplication.

What would it mean to find the mean of observations when $\mathcal{X}, \Theta \neq \mathbb{R}^d$?

Reframe the mean as a minimizer of a squared distance:

$$\bar{X}_n = \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n (X_i - \theta)^2 = \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n d(X_i, \theta)^2$$

Fréchet means

Definition: (Fréchet, 1948) Let (S, d) be a metric space and P a probability measure on S :

- ▶ *Fréchet function:* $F(p) = \int_S d^2(p, x) dP(x), \quad p \in S.$
- ▶ *Fréchet mean set:* $\mu(P) = \arg \min_{p \in S} F(p)$
- ▶ *Fréchet variance:* $V(P) = \inf_{p \in S} F(p).$

Given $X_1, \dots, X_n \stackrel{iid}{\sim} P$

- ▶ *Sample Fréchet function:* $F_n(p) = \frac{1}{n} \sum_i d^2(p, X_i).$
- ▶ *Sample Fréchet mean set:* $\mu(P_n) = \arg \min_{p \in S} F_n(p)$

For $S = \mathbb{R}^d$ with the L^2 distance, $\mu(P) = \{\mathbb{E}X\}$ and $V(P) = \text{traceVar}(X).$

Quick facts about Fréchet means

- ▶ For proper metric spaces, $\mu(P)$ is nonempty
- ▶ Uniqueness is not guaranteed
 - ▶ $M = S^1$, $d = \text{arc-length}$, and P the uniform distribution.
Then $\mu(P) = S^1$.
 - ▶ $M = S^2$, $P = \frac{1}{2}(\delta_N + \delta_S)$ with N, S the North and South Poles.
Then $\mu(P)$ is the equator: $\{p : d(p, N) = \pi/2\}$.
- ▶ Uniqueness requires assumptions on the *support* of P and the *geometry* of M
- ▶ Briefly, we have unique Fréchet means in the following cases
 - ▶ Positive curvature: manifold with $\Delta > 0$ and $\text{supp}(P) \subseteq \bar{B}(o, r)$ with r a function of the injectivity radius and the curvature
 - ▶ Non-positively curved manifolds
 - ▶ Non-positively curved metric spaces

Consistency of Fréchet means

Theorem: (Ziezold 1977) A “strong” LLN for Fréchet means in complete, separable, proper metric spaces:

$$\bigcap_{n \geq 1} \overline{\bigcup_{k \geq n} \mu(P_k)} \subseteq \mu(P) \quad \text{almost surely.}$$

In particular, if the Fréchet mean is unique, $\hat{\mu}_n \xrightarrow{\text{a.s.}} \mu$.

Moving forward, let's think about settings with **unique** Fréchet means.

CLTs in non-Euclidean spaces

Consistency is easy to state on a metric space: a distance becomes small.

Distributional convergence is not so easy: if S is not a vector space, many components of this expression lose meaning:

$$\sqrt{n} (\hat{\mu}_n - \mu)$$

Solution: choose a map from S to $\mathbb{R}^s$, and study the behavior of the Fréchet mean under the map.

A chart at μ

Definition: A *chart at μ* is a measurable $G \subseteq S$ with $\mu \in G$, together with a homeomorphism

$$\phi : G \rightarrow U \subseteq \mathbb{R}^s, \quad U \text{ open},$$

where G carries the subspace topology of S .

We will study

$$\sqrt{n}(\phi(\hat{\mu}_n) - \phi(\mu)).$$

This expression is eventually well-defined by consistency.

Note: S does need not to be a manifold for a chart to exist!

The localised criterion

Write

$$\theta_0 = \phi(\mu), \quad \hat{\theta}_n = \phi(\hat{\mu}_n), \quad m_\theta(x) = d^2(\phi^{-1}(\theta), x) \quad \text{for } \theta \in U,$$

so that

$$H(\theta) = Pm_\theta = F(\phi^{-1}(\theta)), \quad H_n(\theta) = P_n m_\theta = F_n(\phi^{-1}(\theta)).$$

Off the event $\{\hat{\mu}_n \in G\}$ set $\hat{\theta}_n = \theta_0$.

This is starting to look like a Euclidean-valued minimizer θ_0 of a Euclidean-valued function $H(\theta)$!

Conditions for a chart CLT

- (A1) **Uniqueness.** The Fréchet mean μ of P is unique.
- (A2) **Chart and smoothness.** $\mu \in G$, G measurable, and $\phi : G \rightarrow U \subseteq \mathbb{R}^s$ is a homeomorphism onto an open U , G carrying its relative topology; moreover

$$\theta \longmapsto m_\theta(x) = d^2(\phi^{-1}(\theta), x)$$

is twice continuously differentiable on U a.e.- P .

- (A3) **Localisation.** $\Pr(\hat{\mu}_n \in G) \rightarrow 1$.

Conditions for a chart CLT (continued)

(A4) **Moments.** With $D_r = \partial/\partial\theta_r$ and $D_{r,r'} = D_r D_{r'}$,

$$P|D_r m_{\theta_0}|^2 < \infty, \quad P|D_{r,r'} m_{\theta_0}| < \infty, \quad 1 \leq r, r' \leq s.$$

(A5) **Locally uniform L^1 -smoothness of the Hessian.** With

$$u_{r,r'}(\delta; x) = \sup \left\{ |D_{r,r'} m_{\theta}(x) - D_{r,r'} m_{\theta_0}(x)| : \|\theta - \theta_0\| < \delta \right\},$$

$$P u_{r,r'}(\delta; \cdot) \rightarrow 0 \text{ as } \delta \rightarrow 0, \text{ for all } 1 \leq r, r' \leq s.$$

(A6) **Nonsingularity of the Hessian.** $\Lambda = [PD_{r,r'} m_{\theta_0}]_{r,r'}$ is nonsingular.

Lemma: connecting to the Euclidean conditions

Lemma: Assume (A1)–(A3) and $\hat{\mu}_n \xrightarrow{P} \mu$. With $\Theta = U$ and $m_\theta(x) = d^2(\phi^{-1}(\theta), x)$:

- (i) $\theta_0 = \phi(\mu)$ is the unique minimiser of H over U , and $H(\theta_0) = F(\mu) < \infty$;
- (ii) $\hat{\theta}_n \xrightarrow{P} \theta_0$;
- (iii) $\Pr(H_n(\hat{\theta}_n) = \inf_{\theta \in U} H_n(\theta)) \rightarrow 1$.

The omnibus CLT for metric spaces

Theorem: (Bhattacharya–Lin 2017) Under (A1)–(A6),

$$\sqrt{n}(\phi(\hat{\mu}_n) - \phi(\mu)) \xrightarrow{\mathcal{D}} N_s(0, \Lambda^{-1} \Sigma \Lambda^{-1}),$$

where $\Sigma = \text{Var}(\dot{m}_{\theta_0}(X_1))$ and $\Lambda = P\ddot{m}_{\theta_0}$

Proof: We wish to apply Lecture 1's CLT with $m_{\theta}(x) = d^2(\phi^{-1}(\theta), x)$ on $\Theta = U$. The lemma gives us (M1) and (M3). All other conditions are given directly. $\square$

The omnibus CLT for metric spaces

Punchline:

A chart ϕ at a unique Fréchet mean μ turns $\hat{\mu}_n$ into an M-estimator in $\mathbb{R}^s$ for the criterion $m_\theta(x) = d^2(\phi^{-1}(\theta), x)$.

Corollary: intrinsic means

Corollary: (B–L 2017 2.3) Let (M, g) be a complete d -dimensional Riemannian manifold with geodesic distance, let P have a unique intrinsic mean* μ , and suppose P assigns zero mass to some neighbourhood of the cut locus $C(\mu)$. Take $\phi = \text{Log}_\mu$ on a neighbourhood G of μ , $U = \phi(G)$. Under (A4)–(A6), the CLT holds with $s = d$.

The support condition is needed to obtain (A2): $\theta \mapsto d^2(\text{Exp}_\mu \theta, x)$ is smooth as long as $\text{Exp}_\mu \theta \notin C(x)$, and $p \in C(x) \iff x \in C(p)$. Excluding a neighbourhood of $C(\mu)$ gives the required smoothness a.e.- P .

*Fréchet mean set on a manifold when $d =$ geodesic distance.

Case study: The circle

Consider

- ▶ $S = S^1$
- ▶ d = arc length
- ▶ ϕ = signed arc length from μ . $\phi(p) \in (-\pi, \pi]$ and $\phi(\mu) = \theta_0 = 0$.

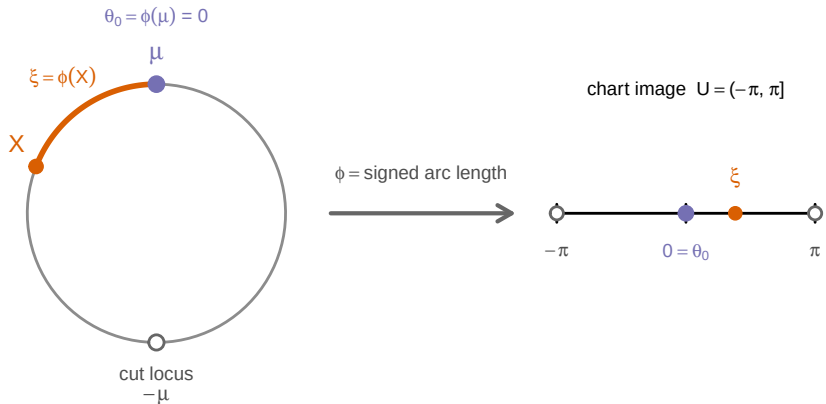
Two points with coordinates θ and $\xi = \phi(x)$ are joined by two arcs; the distance is the shorter one:

$$d(\phi^{-1}(\theta), x) = |\delta(\theta, \xi)|, \quad \delta(\theta, \xi) = \text{wrap}(\theta - \xi) \in (-\pi, \pi],$$

where wrap subtracts a multiple of 2π to land in $(-\pi, \pi]$.

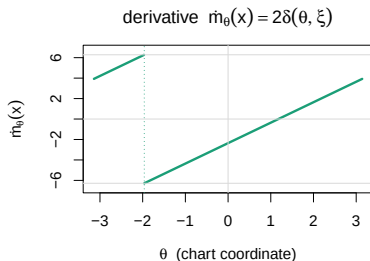
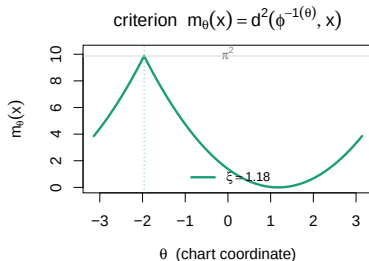
Case study: The circle

The circle S^1 , the mean μ , and the centred chart ϕ



Case study: The circle

The M-estimation criterion is $m_\theta(x) = d^2(\phi^{-1}(\theta), x) = \delta^2(\theta, \xi)$.
For $\xi = \phi(x) = \frac{3\pi}{8} \approx 1.18$, the antipode is at $\frac{-5\pi}{8} \approx -1.96$:



$\dot{m}_\theta(x) = 2\delta(\theta, \xi)$ and $\ddot{m}_\theta(x) = 2$ where they exist. Note that $m_\theta(x)$ **is not** C^1 in θ .

Case study: The circle

Recall (A2): $\theta \mapsto m_\theta(x) = d^2(\phi^{-1}(\theta), x)$ is C^2 a.e.- P .

For a fixed x , $m_\cdot(x)$ is non- C^1 only at $\theta = \phi(-x)$. This point lies in U only when $-x$ is near μ . So (A2) holds on U if the support of P is restricted around the antipode of μ .

i.e., the **support condition** in the corollary.

Case study: The circle

However, mandating a neighbourhood of zero measure is *very restrictive*.

Is it really essential? What's really going on here?

Case study: The circle

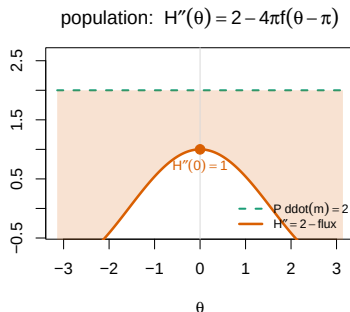
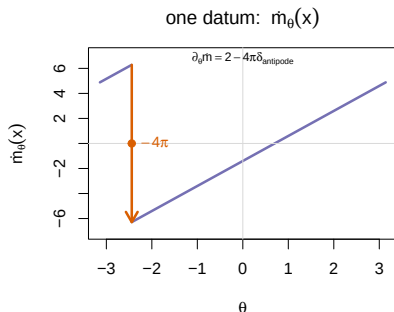
Short answer: inequality of interchanging integration and differentiation:

$$P\ddot{m}_\theta = P \left[\frac{\partial^2}{\partial \theta^2} m_\theta(X) \right] \text{ versus } H''(\theta) = \frac{\partial^2}{\partial \theta^2} P m_\theta(X) = \frac{\partial}{\partial \theta} P \dot{m}_\theta(X)$$

In our circle example, $P\ddot{m}_\theta = 2$ (a.e.) but $H''(\theta) = 2 - 4\pi f(\theta - \pi)!$
Why?

Case study: The circle

Each datum generates a “drop” in $\dot{m}_{\theta_0}$ of -4π . Averaging over these drops lowers $\partial_{\theta} P \dot{m}_{\theta_0}$ compared to $P \ddot{m}_{\theta_0}$. This difference turns out to be $4\pi f(\theta - \pi)$. So, $P \ddot{m}_{\theta}$ and $H''(\theta)$ coincide at $\theta_0 = 0$ if $f(-\pi) = 0$.



Case study: The circle

So, we don't need to restrict a whole neighbourhood! We just need to replace the “wrong” variance $\Lambda = P\ddot{m}_{\theta_0}$ with the “right” variance $V_0 = H''(\theta_0)$.

Where does this “right” variance come from?

Case study: The circle

Consider the estimating functions $\Psi(\theta) = P\dot{m}_\theta = \nabla H(\theta)$ and $\Psi_n(\theta) = P_n\dot{m}_\theta$. Recall $\Psi_n(\hat{\theta}_n) = 0$ and $\Psi(\theta_0) = 0$.

$$0 = \Psi_n(\hat{\theta}_n) = [\Psi_n(\hat{\theta}_n) - \Psi(\hat{\theta}_n)] + [\Psi(\hat{\theta}_n) - \Psi(\theta_0)].$$

► Population term:

$$\Psi(\hat{\theta}_n) - \Psi(\theta_0) = \nabla^2 H(\theta_0)(\hat{\theta}_n - \theta_0) + o_p(\|\hat{\theta}_n - \theta_0\|);$$

► Empirical process term: $\Psi_n(\hat{\theta}_n) - \Psi(\hat{\theta}_n) = (P_n - P)\dot{m}_{\hat{\theta}_n}$

$$(P_n - P)\dot{m}_{\hat{\theta}_n} \stackrel{?}{=} (P_n - P)\dot{m}_{\theta_0} + o_p(n^{-1/2}) = P_n\dot{m}_{\theta_0} + o_p(n^{-1/2}).$$

If “?” holds, then

$$\begin{aligned} 0 &= \nabla^2 H(\theta_0)(\hat{\theta}_n - \theta_0) + o_p(1) + P_n\dot{m}_{\theta_0} + o_p(n^{-1/2}) \\ \implies \sqrt{n}(\hat{\theta}_n - \theta_0) &= -\nabla^2 H(\theta_0)^{-1} \sqrt{n} P_n\dot{m}_{\theta_0} + o_p(1). \end{aligned}$$

Case study: The circle

$(P_n - P)\dot{m}_{\hat{\theta}_n} = (P_n - P)\dot{m}_{\theta_0} + o_p(n^{-1/2})$ is not automatic!

We have a random measure P_n evaluated at a random point $\hat{\theta}_n$ – and both are built from the *same* data.

When will $\sqrt{n}(P_n - P)(\dot{m}_{\hat{\theta}_n} - \dot{m}_{\theta_0}) = o_p(1)$?

When $\{\dot{m}_\theta : \theta \in B_\delta\}$ is “simple enough.”

Case study: The circle

There's a whole theory of what functions are “simple enough” (aka *Donsker*). Here are some sufficient conditions:

- ▶ Lipschitz in θ : $|\dot{m}_{\theta_1}(x) - \dot{m}_{\theta_2}(x)| \leq \dot{h}(x) \|\theta_1 - \theta_2\|$ with $P\dot{h}^2 < \infty$
- ▶ Finite VC complexity: a measure of complexity of classes of functions

Our $\dot{m}_\theta$ is not that complex; it has finite VC complexity; so our $\dot{m}_\theta$ is Donsker and

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\nabla^2 H(\theta_0)^{-1} \sqrt{n} P_n \dot{m}_{\theta_0} + o_p(1).$$

Case study: The circle

Theorem: (Hotz & Huckemann 2015) For continuous f with support on S^1 , if $f(-\pi) < \frac{1}{2\pi}$,

$$\sqrt{n} (\phi(\hat{\mu}_n) - \phi(\mu)) \xrightarrow{\mathcal{D}} N\left(0, \frac{\text{Var}(\xi)}{(1 - 2\pi f(-\pi))^2}\right).$$

i.e., “if the distribution at the antipode is locally less than uniform”, we have a typical $n^{1/2}$ CLT.

This is “just” $V_0^{-1}\Sigma V_0^{-1}$ with $\Sigma = \text{Var}(\dot{m}_{\theta_0}(X)) = \text{Var}(2\delta(0, \xi)) = 4\text{Var}(\xi)$ and $V_0 = H''(\theta_0) = 2(1 - 2\pi f(-\pi))$.

What happens as $f(-\pi) \uparrow \frac{1}{2\pi}$? Or, in a general setting, when $V_0 = H''(\theta_0)$ is singular?

Singular V_0 and smeariness

When $V_0 = H''(\theta_0)$ is singular, we no longer have $\sqrt{n}$ rates!

This can't happen for Euclidean or nonpositively curved spaces: it can happen for positively curved spaces when the Fréchet function flattens at its minimum.

Singular Λ and smeariness

Eltzner & Huckemann give sufficient conditions for a new limiting law for Fréchet means on manifolds.

The conditions involve a sufficiently smooth population Fréchet function; and a well-behaved distance function.

Singular Λ and smeariness

Theorem: (Eltzner & Huckemann 2019): Suppose

$$F(x) = F(0) + \sum_{j=1}^m T_j |(Rx)_j|^r + o(\|x\|^r)$$

near the mean, with $r \geq 2$ and $T_j \neq 0$. Write $k = r - 2$ and $x_n = \exp_{\mu}^{-1}(\hat{\mu}_n)$. Then, componentwise,

$$n^{\frac{1}{2(k+1)}} (Rx_n)_j \xrightarrow{\mathcal{D}} H_j = \text{sign}(G_j) |G_j|^{\frac{1}{k+1}} \quad \begin{aligned} \tilde{\Lambda} &= r T, \\ \tilde{\Sigma} &= \text{Cov}(\text{grad}_x \rho(x, X)|_{x=0}), \end{aligned}$$

for $G \sim \mathcal{N}_m(0, \tilde{\Lambda}^{-1} \tilde{\Sigma} \tilde{\Lambda}^{-1})$.

Singular Λ and smeariness

If the population Fréchet function behaves like $|x|^r$ near the minimum rather than $|x|^2$, then with $k = r - 2$:

$$n^{1/(2(k+1))}(\hat{\mu}_n - \mu) \xrightarrow{\mathcal{D}} H$$

for $H = \text{signed } (k+1)\text{-th root of a Normal random variable.}$

Singular Λ and smeariness

Definition: (Eltzner & Huckemann 2019) For $k > -1$, a sequence of random vectors Z_n are *k-th order smeary* if $n^{1/(2(k+1))}Z_n$ has a nontrivial limiting distribution.

A consistent $\hat{\mu}_n \xrightarrow{P} \mu$ on an m -manifold is *k-smeary* if $\phi(\hat{\mu}_n) - \phi(\mu)$ is *k-smeary*.

Usually, we see non-smeary ($k = 0$) limiting laws: averages, minimizers of smooth functions, ALEs. . .

Singular Λ and smeariness

Smeariness is due to a degenerate Hessian of the Fréchet function, and requires mass at cut loci.

It cannot occur in $\mathbb{R}^d$, and it cannot occur on a NPC space.

But, it's cool!

Summary

- ▶ A chart at a unique Fréchet mean turns $\hat{\mu}_n$ into an M-estimator in $\mathbb{R}^s$ for the criterion $m_\theta(x) = d^2(\phi^{-1}(\theta), x)$.
- ▶ Under (A1)–(A6) the M-estimator CLT applies in the chart: rate $\sqrt{n}$, Gaussian limit, variance $\Lambda^{-1}C\Lambda^{-1}$ with $\Lambda = P\ddot{m}_{\theta_0}$ and $C = \text{Var}(\dot{m}_{\theta_0}(X))$.
- ▶ Intrinsic means: (A2) needs a support condition at the cut locus
- ▶ If $\nabla^2 H(\theta_0)$ is singular and we have positive curvature, we get *smeariness*

References

- ▶ Ziezold (1977). *On expected figures and a strong law of large numbers for random elements in quasi-metric spaces.*
- ▶ Bhattacharya & Patrangenaru (2003, 2005). *Large sample theory of intrinsic and extrinsic sample means on manifolds I, II.* Ann. Statist.
- ▶ Bhattacharya & Lin (2017). *Omnibus CLTs for Fréchet means and nonparametric inference on non-Euclidean spaces.* Proc. Amer. Math. Soc.
- ▶ Kendall & Le (2011). *Limit theorems for empirical Fréchet means of independent and non-identically distributed manifold-valued random variables.*
- ▶ Sturm (2003). *Probability measures on metric spaces of nonpositive curvature.*

References

- ▶ Hotz & Huckemann (2015). *Intrinsic means on the circle: uniqueness, locus and asymptotics*. Ann. Inst. Statist. Math.
- ▶ Hotz et al. (2013). *Sticky central limit theorems on open books*. Ann. Appl. Probab.
- ▶ Eltzner & Huckemann (2019). *A smeary central limit theorem for manifolds with application to high-dimensional spheres*. Ann. Statist.
- ▶ van der Vaart (1998), Ch. 5, 19; van der Vaart & Wellner (1996), Ch. 2.7, 3.2.

Fréchet means

The omnibus CLT

Assumption failure and smeariness

A blank slide!

Asymptotic statistics on stratified spaces

Lecture 2: CLTs on metric spaces and manifolds

Slides available at github.com/statdivlab/asymptotics-stratified-spaces

Amy Willis

September 2026